



Synchronous Reluctance Motor

KSB SuPremE

Type Series Booklet



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Type Series Booklet KSB SuPremE

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Drives

Synchronous Reluctance Motor

KSB SuPremE



 The product illustrated as an example may include options incurring a surcharge.

Main applications

- Dry-installed pumps, in particular those that accumulate a high number of operating hours per year and have to accommodate fluctuating loads
- Rotating equipment

Design details

- Magnetless synchronous reluctance motor¹⁾
- Without rotor position sensors
- Rotor with air gaps (in accordance with US patent No. 5818140)
- Mounting points to IEC 60072-1:2022
- Envelope dimensions to DIN VDE 42673-4:2011-07
- Self-cooling (design: TEFC)
- Shaft centreline height 71 to 225 mm
- Rated power 0.55 kW to 45 kW

Design

KSB SuPremE C1/D1:

- With terminal box for connecting to PumpDrive 2 or PumpDrive R for mounting on walls and in control cabinets

KSB SuPremE C2/D2:

- Equipped for being fitted with a motor-mounted PumpDrive 2

Connections

Flange type, designations to IEC 60072-1:2022

Without flange:

- Type of construction IM B3 (standard), with foot, shaft centreline height 71 to 225 mm
- Type of construction IM B6/B7/B8/V5/V6, with foot, shaft centreline height 71 to 225 mm

Flange with clearance holes (FF):

- Type of construction IM V15 (standard), with foot, shaft centreline height 71 to 225 mm
- Type of construction IM V35 (not for special design for Movitec), with foot, shaft centreline height 71 to 225 mm
- Type of construction IM B35, with foot, shaft centreline height 71 to 225 mm
- Type of construction IM V1 (standard), without foot, shaft centreline height 71 to 225 mm
- Type of construction IM V3 (not for special design for Movitec), without foot, shaft centreline height 71 to 225 mm
- Type of construction IM B5, without foot, shaft centreline height 71 to 225 mm

Flange with tapped holes (FT), special design for Movitec:

- Type of construction IM V18 (standard), without foot, shaft centreline height ≤ 132 mm
- Type of construction IM B14/V19, without foot, shaft centreline height ≤ 132 mm

¹ Exception: Motor sizes 0.55 kW / 0.75 kW with 1500 rpm are designed with permanent magnets.

Bearings

Table 1: Non-insulated bearings used

Motor size	Generation	Type of construction B3 / V1 / V15		Special design Movitec	
		Drive end	Non-drive end	Drive end	Non-drive end
71M	B	E2.6203-2ZC3	E2.6203-2ZC3	E2.6203-2ZC3	E2.6203-2ZC3
71M	C	E2.6202-2ZC3	E2.6202-2ZC3	E2.6203-2ZC3	E2.6202-2ZC3
80M	B	E2.6204-2ZC3	E2.6204-2ZC3	E2.6204-2ZC3	E2.6204-2ZC3
80M	C	E2.6204-2ZC3	E2.6204-2ZC3	E2.6204-2ZC3	E2.6204-2ZC3
80M	D	E2.6204-2ZC3	E2.6204-2ZC3	-	-
90S/L	B	E2.6205-2ZC3	E2.6205-2ZC3	E2.6305-2ZC3	E2.6305-2ZC3
90S/L	C	E2.6205-2ZC3	E2.6205-2ZC3	E2.6305-2ZC3	E2.6205-2ZC3
100L	B	E2.6206-2ZC3	E2.6206-2ZC3	E2.6306-2ZC3	E2.6306-2ZC3
100L	C	E2.6206-2ZC3	E2.6206-2ZC3	E2.6306-2ZC3	E2.6206-2ZC3
100L	D	E2.6206-2ZC3	E2.6206-2ZC3	-	-
112M	B	E2.6306-2ZC3	E2.6306-2ZC3	E2.6306-2ZC3	E2.6306-2ZC3
112M	C	E2.6206-2ZC3	E2.6206-2ZC3	E2.6306-2ZC3	E2.6206-2ZC3
112M	D	E2.6206-2ZC3	E2.6206-2ZC3	-	-
132M/S	B	E2.6308-2ZC3	E2.6308-2ZC3	E2.6308-2ZC3	E2.6308-2ZC3
132M/S	C	E2.6208-2ZC3	E2.6208-2ZC3	E2.6308-2ZC3	E2.6208-2ZC3
160M/L	B	E2.6309-2ZC3	E2.6309-2ZC3	7309	E2.6309-2ZC3
160M/L	C	E2.6309-2ZC3	E2.6309-2ZC3	7309	E2.6309-2ZC3
180M/L	B	E2.6310-2ZC3	E2.6310-2ZC3	7311	E2.6310-2ZC3
180M/L	C	6210-ZJC3	6210-ZJC3	7310	6210-ZJC3
180M/L ²⁾	D	6210-J2ZC3	6210-J2ZC3	7310 BEP	6210-J2ZC3
200L	B	6313-2ZC3	6313-2ZC3	7312	6313-2ZC3
200L	C	6212-ZJC3	6212-ZJC3	7312	6212-ZJC3
200L ²⁾	D	6212-J2ZC3	6212-J2ZC3	7312 BEP	6212-J2ZC3
225S/M	B	6314-2ZC3	6314-2ZC3	7313	6314-2ZC3
225S/M	C	6213-ZJC3	6213-ZJC3	7313	6213-ZJC3
225S/M	D	6213-JZC3	6213-JZC3	7313 BEP	6213-JZC3

Table 2: Insulated bearings used

Motor size	Generation	Type of construction B3 / V1 / V15		Special design Movitec	
		Drive end	Non-drive end	Drive end	Non-drive end
180M/L	D	6210-JZC3	6210-2RS1 HC5 C3WT	7310 BEP	6210-2RS1 HC5 C3WT
200L	D	6212-JZC3	6213-JZC3 ³⁾⁴⁾	7312 BEP	6213-JZC3 ³⁾⁴⁾
225S/M	D	6213-JZC3	6213-JZC3 ³⁾	7313 BEP	6213-JZC3 ³⁾

2 From serial number UC/UD 2111: changed from Z bearings to 2Z bearings

3 Insulated bearing seat (shaft)

4 From production date 25/01: changeover from 6212-2RS1 HC5 C3WT to 6213-JZC3

Designation

Table 3: Designation example

Position																													
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
-	2	-	4	5	,	0	-	2	2	5	M	-	B	W	A	7	F	3	N	R	S	D	W	F	Z	W	K	S	F

Table 4: Designation key

Position	Code	Description
1-2	Nominal speed n [rpm]	
	2	3000
	4	1500
4-7	Motor rating P _N [kW]	
	0,55	0,55
	...	...
9-12	Motor size	
	71M	Shaft centreline height [mm]
	...	...
14	Enclosure	
	B	IP55 or IP40/55
	...	...
15	Type of protection	
	W	Non-explosion-proof
	...	...
16	Voltage	
	A / P	3~, AC, 220 VΔ, 380 VY, 50 Hz
	...	...
17	Efficiency class	
	7	IE4 / IE5 (NEMA Super Premium / Ultra Premium)
	...	...
18	Thermal class	
	F	Thermal class 155 (F)
	...	...
19	Motor protection / winding protection	
	3	3 PTC thermistors
	...	...
20	Direction of rotation	
	N	Clockwise / counter-clockwise (bi-directional)
	...	...
21	Position of terminal box	
	T	Terminal box on top
	N	Not defined
22	Feet mounting	
	S	Feet attached by bolts
	W	Without feet
23	Position of fixed bearing	
	D	Fixed bearing, drive end
	C	Fixed bearing, drive end, reinforced
24	Protective roof	
	W	Without protective roof
	...	...
25	Motor flange	
	F	IEC 60072, type FF
	T	IEC 60072, type FT
26	Operation on a frequency inverter	
	Z	Compulsory operation on a frequency inverter
	...	...
27	Approval	
	W	Without approvals
	...	...

Position	Code	Description
28-30	Manufacturer	
	KSB	KSB SuPremE B, primed
	KSG	KSB SuPremE B, vermillion, RAL 2002
	KSH	KSB SuPremE B, ultramarine blue, RAL 5002
	SCD	KSB SuPremE C, ultramarine blue, RAL 5002
	SDD	KSB SuPremE D, ultramarine blue, RAL 5002

Speed / frequency

Table 5: Available speeds and frequencies

Generation	Rated speed	Maximum speed	Rated frequency	Maximum frequency
	[rpm]	[rpm]	[Hz]	[Hz]
B / C / D	1500	2100	50	70
D (0,55 kW / 0,75 kW)	1500	2100	125	175
B / C / D	3000	4200	100	140

Product benefits

- Energy-efficient operation thanks to high efficiency at the rated load point
- The efficiency of the motor also exceeds 95 % of the nominal efficiency when the motor runs at 25 % of its nominal power on a quadratic torque-speed curve.
- Fully interchangeable with IE2 motors thanks to compliance with DIN EN IEC 60072-1:2022 and envelope dimensions in accordance with DIN V 42673-4
- Low noise emissions and torque ripple due to use of rotor with air gaps (in accordance with US Patent No. 5818140)
- Sturdy design, no additional sensors
- Lower rotor temperature increases service life of bearings.
- Sustainable and environmentally friendly because no magnets based on "rare earth elements" such as NdFeB are used
- Compact variable speed drive when combined with motor-mounted PumpDrive 2 / PumpDrive 2 Eco

Product information

Product information as per Regulation No. 1907/2006 (REACH)

For information as per European chemicals regulation (EC) No. 1907/2006 (REACH) see <https://www.ksb.com/en-global/company/corporate-responsibility/reach>.

Noise characteristics

Table 6: Sound pressure level L_{pA}

Rated speed	Rated power	Motor size	Sound pressure level L_{pA} ⁵⁾
[rpm]	[kW]		[dB (A)]
1500	0,55	80M	60
1500	0,75	80M	60
1500	1,1	90S	60
1500	1,5	90L	60
1500	2,2	100L	60
1500	3	100L	60
1500	4	112M	61
1500	5,5	132S	61
1500	7,5	132M	61
1500	11	160M	61
1500	15	160L	61
1500	18,5	180M	63
1500	22	180L	63
1500	30	200L	64
1500	37	225S	64
1500	45	225M	64
3000	0,55	71M	70
3000	0,75	80M	70
3000	1,1	80M	70
3000	1,5	90S	70
3000	2,2	90L	70
3000	3	100L	71
3000	4	112M	71
3000	5,5	132S	71
3000	7,5	132S	71
3000	11	160M	71
3000	15	160M	71
3000	18,5	160L	72
3000	22	180M	72
3000	30	200L	72
3000	37	200L	72
3000	45	225M	72

5 Measured at idle with rated speed to IEC 60034-9

Versions

Versions with non-insulated bearings

i Motors of **types of construction IM B3, V1 and V15** are standard motor designs with a standardised shaft end (IEC 60072) and standard bearings.

i Motors in **special design for Movitec** are fitted with the motor options required for the Movitec pump type series. They are designed with a plain shaft end and, depending on the size, with a flange or reinforced bearings.

Table 7: Selection table

Type of construction/ special design	Efficiency class	Rated speed	Rated power	Motor	KSB SuPremE + terminal box	KSB SuPremE + mounting plate for PumpDrive 2 / PumpDrive 2 Eco	KSB SuPremE + PumpDrive 2	KSB SuPremE + PumpDrive 2 Eco
					Blue (RAL 5002)	Blue (RAL 5002)	Blue (RAL 5002)	Blue (RAL 5002)
		[rpm]	[kW]		Mat. No.	Mat. No.	Mat. No.	Mat. No.
IM B3	IE5	1500	0,55 ⁶⁾	080M	05010022	05010025	01656802	01656918
IM B3	IE5	1500	0,75 ⁶⁾	080M	05010037	05010040	01656803	01656919
IM B3	IE4	1500	1,10	090S	05231330	05231326	.. ⁷⁾	.. ⁷⁾
IM B3	IE4	1500	1,50	090L	05231329	05231325	.. ⁷⁾	.. ⁷⁾
IM B3	IE5	1500	2,20	100L	01677991	01639912	01656806	01656922
IM B3	IE5	1500	3,00	100L	01677990	01639911	01656807	01656923
IM B3	IE5	1500	4,00	112M	01677989	01639910	01656808	01656924
IM B3	IE4	1500	5,50	132S	05231402	05231358	.. ⁷⁾	.. ⁷⁾
IM B3	IE5	1500	7,50	132M	01677987	01639908	01656810	01656926
IM B3	IE5	1500	11,00	160M	01677916	01639907	01656811	01656927
IM B3	IE5	1500	15,00	160L	01677915	01639906	01656812	.. ⁷⁾
IM B3	IE4	1500	18,50	180M	01677914	01639905	01656813	.. ⁷⁾
IM B3	IE4	1500	22,00	180L	01677913	01639904	01656814	.. ⁷⁾
IM B3	IE4	1500	30,00	200L	01677912	01639903	01656815	.. ⁷⁾
IM B3	IE4	1500	37,00	225S	01677911	01639902	01656816	.. ⁷⁾
IM B3	IE4	1500	45,00	225M	01677910	01639901	01656817	.. ⁷⁾
IM B3	IE5	3000	0,75	080M	01677903	01639899	01656819	01656929
IM B3	IE5	3000	1,10	080M	01677904	01639898	01656820	01656930
IM B3	IE5	3000	1,50	090S	01677905	01639897	01656821	01656931
IM B3	IE5	3000	2,20	090L	01677906	01639896	01656822	01656932
IM B3	IE5	3000	3,00	100L	01677927	01639895	01656823	01656933
IM B3	IE5	3000	4,00	112M	01677928	01639894	01656824	01656934
IM B3	IE5	3000	5,50	132S	01677929	01639893	01656825	01656935
IM B3	IE5	3000	7,50	132S	01677930	01639892	01656826	01656936
IM B3	IE5	3000	11,00	160M	01677931	01639891	01656827	01656937
IM B3	IE5	3000	15,00	160M	01677932	01639890	01656828	.. ⁷⁾
IM B3	IE5	3000	18,50	160L	01677933	01639889	01656829	.. ⁷⁾
IM B3	IE4	3000	22,00	180M	01677934	01639888	01656830	.. ⁷⁾
IM B3	IE4	3000	30,00	200L	01677935	01639887	01656831	.. ⁷⁾
IM B3	IE4	3000	37,00	200L	01677936	01639886	01656832	.. ⁷⁾
IM B3	IE4	3000	45,00	225M	01677937	01639885	01656833	.. ⁷⁾
IM V1	IE5	1500	0,55 ⁶⁾	080M	05010023	05010026	01656834	01656938
IM V1	IE5	1500	0,75 ⁶⁾	080M	05010038	05010041	01656835	01656939
IM V1	IE4	1500	1,10	090S	05231332	05231320	.. ⁷⁾	.. ⁷⁾
IM V1	IE4	1500	1,50	090L	05231331	05231249	.. ⁷⁾	.. ⁷⁾
IM V1	IE5	1500	2,20	100L	01678006	01639816	01656838	01656942
IM V1	IE5	1500	3,00	100L	01678005	01639815	01656839	01656943
IM V1	IE5	1500	4,00	112M	01678004	01639814	01656840	01656944

6 IE5 motors of sizes 0.55 kW / 0.75 kW with 1500 rpm are designed with permanent magnets; therefore a cogging torque is present when the motor is de-energised.

7 Not available

Type of construction/ special design	Efficiency class	Rated speed	Rated power	Motor	KSB SuPremE + terminal box	KSB SuPremE + mounting plate for PumpDrive 2 / PumpDrive 2 Eco	KSB SuPremE + PumpDrive 2	KSB SuPremE + PumpDrive 2 Eco
					Blue (RAL 5002)	Blue (RAL 5002)	Blue (RAL 5002)	Blue (RAL 5002)
		[rpm]	[kW]		Mat. No.	Mat. No.	Mat. No.	Mat. No.
IM V1	IE4	1500	5,50	132S	05231404	05231352	.. ⁷⁾	.. ⁷⁾
IM V1	IE5	1500	7,50	132M	01678002	01639812	01656842	01656946
IM V1	IE5	1500	11,00	160M	01678001	01639811	01656843	01656947
IM V1	IE5	1500	15,00	160L	01678000	01639810	01656844	.. ⁷⁾
IM V1	IE4	1500	18,50	180M	01677999	01639809	01656845	.. ⁷⁾
IM V1	IE4	1500	22,00	180L	01677998	01639808	01656846	.. ⁷⁾
IM V1	IE4	1500	30,00	200L	01677997	01639807	01656847	.. ⁷⁾
IM V1	IE4	1500	37,00	225S	01677996	01639806	01656848	.. ⁷⁾
IM V1	IE4	1500	45,00	225M	01677995	01639805	01656849	.. ⁷⁾
IM V1	IE5	3000	0,55	071M	01677953	01639804	01656850	01656948
IM V1	IE5	3000	0,75	080M	01677952	01639803	01656851	01656949
IM V1	IE5	3000	1,10	080M	01677951	01639802	01656852	01656950
IM V1	IE5	3000	1,50	090S	01677950	01639801	01656853	01656851
IM V1	IE5	3000	2,20	090L	01677949	01639800	01656854	01656852
IM V1	IE5	3000	3,00	100L	01677948	01639799	01656855	01656953
IM V1	IE5	3000	4,00	112M	01677947	01639798	01656856	01656954
IM V1	IE5	3000	5,50	132S	01677946	01639797	01656857	01656955
IM V1	IE5	3000	7,50	132S	01677945	01639786	01656858	01656956
IM V1	IE5	3000	11,00	160M	01677944	01639785	01656859	01656957
IM V1	IE5	3000	15,00	160M	01677943	01639784	01656860	.. ⁷⁾
IM V1	IE5	3000	18,50	160L	01677942	01639783	01656861	.. ⁷⁾
IM V1	IE4	3000	22,00	180M	01677941	01639782	01656862	.. ⁷⁾
IM V1	IE4	3000	30,00	200L	01677940	01639781	01656863	.. ⁷⁾
IM V1	IE4	3000	37,00	200L	01677939	01639780	01656864	.. ⁷⁾
IM V1	IE4	3000	45,00	225M	01677938	01639779	01656865	.. ⁷⁾
IM V15	IE5	1500	0,55 ⁶⁾	080M	05010024	05010027	01656866	01656958
IM V15	IE5	1500	0,75 ⁶⁾	080M	05010039	05010042	01656867	01656959
IM V15	IE4	1500	1,10	090S	05231334	05231322	.. ⁷⁾	.. ⁷⁾
IM V15	IE4	1500	1,50	090L	05231333	05231321	.. ⁷⁾	.. ⁷⁾
IM V15	IE5	1500	2,20	100L	01678022	01639848	01656870	01656962
IM V15	IE5	1500	3,00	100L	01678021	01639847	01656871	01656963
IM V15	IE5	1500	4,00	112M	01678020	01639846	01656872	01656964
IM V15	IE4	1500	5,50	132S	05231406	05231354	.. ⁷⁾	.. ⁷⁾
IM V15	IE5	1500	7,50	132M	01678018	01639844	01656874	01656966
IM V15	IE5	1500	11,00	160M	01678017	01639843	01656875	01656967
IM V15	IE5	1500	15,00	160L	01678016	01639842	01656876	.. ⁷⁾
IM V15	IE4	1500	18,50	180M	01678015	01639841	01656877	.. ⁷⁾
IM V15	IE4	1500	22,00	180L	01678014	01639840	01656878	.. ⁷⁾
IM V15	IE4	1500	30,00	200L	01678013	01639839	01656879	.. ⁷⁾
IM V15	IE4	1500	37,00	225S	01678012	01639838	01656880	.. ⁷⁾
IM V15	IE4	1500	45,00	225M	01678011	01639837	01656881	.. ⁷⁾
IM V15	IE5	3000	0,75	080M	01677963	01639835	01656883	01656969
IM V15	IE5	3000	1,10	080M	01677962	01639834	01656884	01656970
IM V15	IE5	3000	1,50	090S	01677961	01639833	01656885	01656971
IM V15	IE5	3000	2,20	090L	01677960	01639832	01656886	01656972
IM V15	IE5	3000	3,00	100L	01677959	01639831	01656887	01656973
IM V15	IE5	3000	4,00	112M	01677958	01639830	01656888	01656974
IM V15	IE5	3000	5,50	132S	01677957	01639829	01656889	01656975
IM V15	IE5	3000	7,50	132S	01677926	01639828	01656890	01656976
IM V15	IE5	3000	11,00	160M	01677925	01639827	01656891	01656977
IM V15	IE5	3000	15,00	160M	01677924	01639826	01656892	.. ⁷⁾

Type of construction/ special design	Efficiency class	Rated speed	Rated power	Motor	KSB SuPremE + terminal box	KSB SuPremE + mounting plate for PumpDrive 2 / PumpDrive 2 Eco	KSB SuPremE + PumpDrive 2	KSB SuPremE + PumpDrive 2 Eco
		[rpm]	[kW]		Blue (RAL 5002)	Blue (RAL 5002)	Blue (RAL 5002)	Blue (RAL 5002)
					Mat. No.	Mat. No.	Mat. No.	Mat. No.
IM V15	IE5	3000	18,50	160L	01677923	01639825	01656893	.. ⁷⁾
IM V15	IE4	3000	22,00	180M	01677922	01639824	01656894	.. ⁷⁾
IM V15	IE4	3000	30,00	200L	01677921	01639823	01656895	.. ⁷⁾
IM V15	IE4	3000	37,00	200L	01677920	01639822	01656896	.. ⁷⁾
IM V15	IE4	3000	45,00	225M	01677919	01639821	01656897	.. ⁷⁾
Movitec	IE5	3000	0,55	071M	01677892	01639859	01656786	01656908
Movitec	IE5	3000	0,75	080M	01677891	01639858	01656787	01656909
Movitec	IE5	3000	1,10	080M	01677890	01639857	01656788	01656910
Movitec	IE5	3000	1,50	090S	01677889	01639856	01656789	01656911
Movitec	IE5	3000	2,20	090L	01677888	01639855	01656790	01656912
Movitec	IE5	3000	3,00	100L	01677887	01639854	01656791	01656913
Movitec	IE5	3000	4,00	112M	01677766	01639853	01656792	01656914
Movitec	IE5	3000	5,50	132S	01677894	01639868	01656793	01656915
Movitec	IE5	3000	7,50	132S	01677893	01639867	01656794	01656916
Movitec	IE5	3000	11,00	160M	01677901	01639877	01656795	01656917
Movitec	IE5	3000	15,00	160M	01677900	01639876	01656796	.. ⁷⁾
Movitec	IE5	3000	18,50	160L	01677899	01639875	01656797	.. ⁷⁾
Movitec	IE4	3000	22,00	180M	01677898	01639874	01656798	.. ⁷⁾
Movitec	IE4	3000	30,00	200L	01677897	01639873	01656799	.. ⁷⁾
Movitec	IE4	3000	37,00	200L	01677896	01639872	01656800	.. ⁷⁾
Movitec	IE4	3000	45,00	225M	01677895	01639871	01656801	.. ⁷⁾

Versions with insulated bearings

i Motors of **types of construction IM B3, V1 and V15** are standard motor designs with a standardised shaft end (IEC 60072) and standard bearings.

i Motors in **special design for Movitec** are fitted with the motor options required for the Movitec pump type series. They are designed with a plain shaft end and, depending on the size, with a flange or reinforced bearings.

Table 8: Selection table

Type of construction/ special design	Efficiency class	Rated speed	Rated power	Motor	KSB SuPremE + terminal box	KSB SuPremE + mounting plate for PumpDrive 2 / PumpDrive 2 Eco
		[rpm]	[kW]		Blue (RAL 5002)	Blue (RAL 5002)
					Mat. No.	Mat. No.
IM B3	IE4	1500	18,50	180M	05108221	05108318
IM B3	IE4	1500	22,00	180L	05108220	05108317
IM B3	IE4	1500	30,00	200L	05108089	05108316
IM B3	IE4	1500	37,00	225S	05108088	05108315
IM B3	IE4	1500	45,00	225M	05108087	05108314
IM B3	IE4	3000	22,00	180M	05108226	05108313
IM B3	IE4	3000	30,00	200L	05108227	05108312
IM B3	IE4	3000	37,00	200L	05108228	05108311
IM B3	IE4	3000	45,00	225M	05108229	05108310
IM V1	IE4	1500	18,50	180M	05108243	05108291
IM V1	IE4	1500	22,00	180L	05108242	05108290
IM V1	IE4	1500	30,00	200L	05108241	05108079
IM V1	IE4	1500	37,00	225S	05108240	05108078
IM V1	IE4	1500	45,00	225M	05108239	05108077
IM V1	IE4	3000	22,00	180M	05108233	05108076
IM V1	IE4	3000	30,00	200L	05108232	05108075
IM V1	IE4	3000	37,00	200L	05108231	05108074
IM V1	IE4	3000	45,00	225M	05108230	05108073
IM V15	IE4	1500	18,50	180M	05108248	05108300
IM V15	IE4	1500	22,00	180L	05108247	05108299
IM V15	IE4	1500	30,00	200L	05108246	05108298
IM V15	IE4	1500	37,00	225S	05108245	05108297
IM V15	IE4	1500	45,00	225M	05108244	05108296
IM V15	IE4	3000	22,00	180M	05108225	05108295
IM V15	IE4	3000	30,00	200L	05108224	05108294
IM V15	IE4	3000	37,00	200L	05108223	05108293
IM V15	IE4	3000	45,00	225M	05108222	05108292
Movitec	IE4	1500	18,50	180M	05108238	05108309
Movitec	IE4	1500	22,00	180L	05108237	05108308
Movitec	IE4	1500	30,00	200L	05108236	05108307
Movitec	IE4	1500	37,00	225S	05108235	05108306
Movitec	IE4	1500	45,00	225M	05108234	05108305
Movitec	IE4	3000	22,00	180M	05108086	05108304
Movitec	IE4	3000	30,00	200L	05108085	05108303
Movitec	IE4	3000	37,00	200L	05108084	05108302
Movitec	IE4	3000	45,00	225M	05108083	05108301

Technical data
Electrical data for all motors depending on rated power and rated speed
Table 9: Technical data

Characteristic		Value
Ambient temperature		-20 °C to +40 °C
Atmospheric humidity		Maximum of 55 % relative atmospheric humidity at 40 °C ambient temperature
Altitude		≤ 1000 m (without power derating)
Thermal class		F
Locked-rotor torque in [%] of rated torque		
	IC 411 continuous	50 %
	IC 416 continuous	100%
	IC 411/416, 10 seconds max.	110 %
Maximum torque in [%] of rated torque		110 %
Enclosure		
	KSB SuPremE with terminal box	IP55
	KSB SuPremE with adapter plate for PumpDrive 2	IP55 ⁸⁾
Paint coat		RAL 5002 or, if combined with a pump, in the same colour as the pump

Table 10: Electrical data for all motors depending on rated power and rated speed

Rated speed	Rated power	Motor	Generation	Rated voltage	Rated current	Rated torque	Power factor at rated operating point $\cos \varphi$	Moment of inertia of rotor	Weight	Size PumpDrive 2 Eco
[rpm]	[kW]	-	-	[V]	[A]	[Nm]	-	[kgm ²]	[kg]	-
1500	0,55	080M	B/C	350	1,6	3,5	0,67	0,0015	11	A
1500	0,55	80M	D ¹⁰⁾	350	1,3	3,5	0,79	0,0011	8,5	A
1500	0,75	080M	B/C	355	2,1	4,8	0,68	0,0018	13	A
1500	0,75	80M	D ¹⁰⁾	300	2,0	4,8	0,80	0,0014	11,7	A
1500	1,1	090S	B/C/D	360	3,0	7,0	0,67	0,0019	15	A
1500	1,5	090L	B/C/D	365	4,0	9,5	0,67	0,0024	18	A
1500	2,2	100L	B/C	365	5,7	14,0	0,68	0,004	25	B
1500	2,2	100L	D	340	5,9	14,0	0,69	0,0053	26	B
1500	3	100L	B/C	355	7,8	19,1	0,69	0,0046	30	B
1500	3	100L	D	360	7,3	19,1	0,73	0,0051	27	B
1500	4	112M	B/C	360	9,6	25,5	0,73	0,0075	37	B
1500	4	112M	D	360	9,6	25,5	0,72	0,0086	36	B
1500	5,5	132S	B/C/D	350	13,5	35,0	0,73	0,018	45	C
1500	7,5	132M	B/C/D	355	17,6	47,7	0,75	0,026	60	C
1500	11	160M	B/C/D	365	24,2	70,0	0,77	0,051	81	C
1500	15	160L	B/C/D	355	33,1	95,5	0,78	0,063	107	D
1500	18,5	180M	B	350	42,2	117,8	0,77	0,12	151	D
1500	18,5	180M	C	380	42,0	117,8	0,71	0,12	189	D
1500	22	180L	B	365	48,5	140,1	0,76	0,14	175	D
1500	22	180L	C	380	50,0	140,1	0,71	0,13	203	D
1500	30	200L	B	365	65,4	191,0	0,76	0,21	239	D
1500	30	200L	C	380	68,0	191,0	0,71	0,19	246	D
1500	37	225S	B	360	80,9	235,5	0,77	0,34	348	E
1500	37	225S	C	380	81,0	235,5	0,73	0,48	337	E
1500	37	225S	D	380	79,0	236,0	0,75	0,442	340	E
1500	45	225M	B	360	99,3	286,5	0,76	0,38	396	E
1500	45	225M	C	380	98,0	286,5	0,73	0,48	347	E

- 8** With motor-mounted PumpDrive 2
9 No effect on power supply network
10 Permanent magnet motor

Rated speed	Rated power	Motor	Generation	Rated voltage	Rated current	Rated torque	Power factor at rated operating point $\cos \varphi$	Moment of inertia of rotor	Weight	Size PumpDrive 2 Eco
[rpm]	[kW]	-	-	[V]	[A]	[Nm]	-	[kgm ²]	[kg]	-
1500	45	225M	D	380	96,0	286,0	0,75	0,518	343	E
3000	0,55	071M	B/C	350	1,6	1,8	0,68	0,0004	9	A
3000	0,75	080M	B/C	360	2,1	2,4	0,67	0,0013	10	A
3000	1,1	080M	B/C	360	3,0	3,5	0,68	0,0016	12	A
3000	1,5	090S	B/C/D	380	4,1	4,8	0,63	0,017	15	A
3000	2,2	090L	B/C/D	360	5,6	7,0	0,71	0,0025	18	B
3000	3	100L	B/C	360	7,6	9,5	0,70	0,0043	24	B
3000	4	112M	B/C	355	9,4	12,7	0,76	0,0067	35	B
3000	5,5	132S	B/C/D	365	12,5	17,5	0,76	0,015	43	C
3000	7,5	132S	B/C/D	370	16,7	23,9	0,76	0,018	56	C
3000	11	160M	B/C/D	375	23,7	35,0	0,77	0,042	73	C
3000	15	160M	B/C/D	365	32,0	47,7	0,79	0,051	82	D
3000	18,5	160L	B/C/D	370	38,8	58,9	0,79	0,063	102	D
3000	22	180M	B	355	50,7	70,0	0,75	0,12	157	D
3000	22	180M	C	380	50,0	70,0	0,71	0,09	167	D
3000	30	200L	B	365	63,5	95,5	0,79	0,19	222	D
3000	30	200L	C	380	67,0	95,5	0,72	0,17	212	D
3000	37	200L	B	360	77,8	117,8	0,80	0,21	257	E
3000	37	200L	C	380	82,0	117,8	0,72	0,19	257	E
3000	45	225M	B	360	97,0	143,2	0,78	0,34	396	E
3000	45	225M	C	380	100,0	143,2	0,72	0,36	330	E
3000	45	225M	D	380	99,0	143,0	0,73	0,356	305	E

Power loss to DIN EN IEC 61800-9-2:2025

The loss values indicated apply to the power drive system (PDS) consisting of a KSB SuPremE motor and a KSB PumpDrive 2 / PumpDrive 2 Eco frequency inverter with the same rated power.

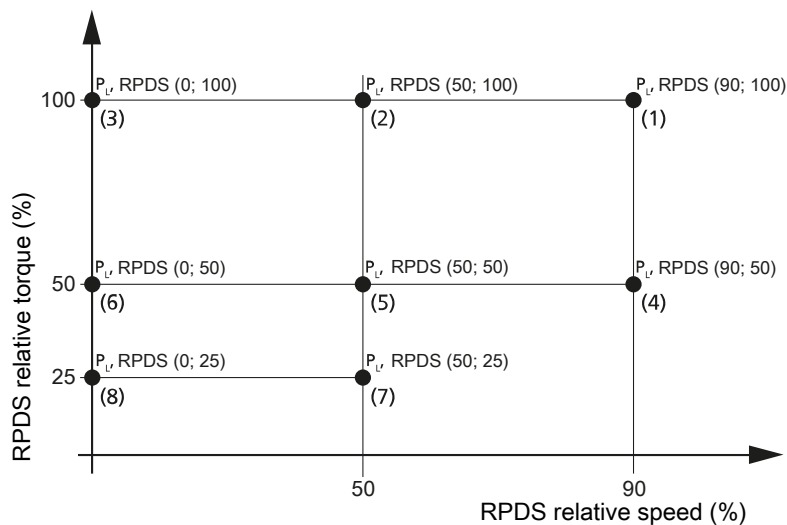


Fig. 1: Reference points to DIN EN IEC 61800-9-2:2025

Table 11: Values for the reference points P_L , RPDS (1) to (8) defined in DIN EN IEC 61800-9-2:2025

P_N [kW]	n [rpm]	IES class	PL compared to the rated power [%]							
			Reference points							
			1	2	3	4	5	6	7	8
0,55 ¹⁾	1500	IES5	25,3	23,2	19,7	15,9	14,1	11,3	9,8	7,7
0,55	1500	IES5	18,9	16,9	19,5	9,8	8,0	10,5	5,8	8,7
0,75 ¹⁾	1500	IES5	20,8	21,7	19,6	15,1	14,1	11,8	11,6	9,3
0,75	1500	IES5	15,2	13,1	16,4	7,5	7,3	10,3	5,2	7,7
1,1 ¹⁾	1500	IES5	19,3	17,9	14,3	12,6	11,7	9,3	9,9	8,3
1,1	1500	IES5	17,3	15,0	14,2	9,7	8,2	7,4	5,0	4,8
1,5 ¹⁾	1500	IES5	17,7	16,6	12,4	9,9	9,0	8,4	5,9	6,2
1,5	1500	IES5	15,3	14,5	14,2	8,5	7,5	7,3	4,7	4,5
2,2	1500	IES5	13,0	11,7	10,8	7,5	6,5	6,1	4,1	3,6
3	1500	IES5	12,9	11,7	10,3	6,6	5,9	5,5	3,7	2,9
4	1500	IES5	11,7	10,8	10,0	6,2	5,5	4,9	3,4	2,1
5,5 ¹⁾	1500	IES5	12,9	11,9	10,3	6,6	5,7	5,0	3,4	3,0
5,5	1500	IES5	11,7	10,7	9,2	6,5	5,4	4,3	2,9	2,6
7,5	1500	IES5	10,3	9,0	8,2	5,0	4,5	4,1	2,4	2,4
11	1500	IES5	9,6	8,7	6,3	5,0	3,9	3,3	2,3	2,0
15	1500	IES5	8,0	7,3	5,3	3,5	3,6	2,9	1,8	1,7
18,5	1500	IES5	7,7	6,2	4,7	4,1	3,1	2,6	2,0	1,6
22	1500	IES5	7,8	6,0	4,7	4,1	3,2	2,5	2,0	1,6
30	1500	IES5	7,4	6,4	5,0	4,2	3,4	2,5	2,0	4,5
37	1500	IES5	7,3	5,9	4,7	3,8	3,0	2,5	1,8	2,7
45	1500	IES5	7,2	5,5	4,4	3,5	2,7	2,4	1,6	1,4
0,55	3000	IES5	22,9	17,1	14,9	15,5	11,1	9,4	8,3	6,8
0,75	3000	IES5	20,6	15,2	13,0	14,5	11,6	9,8	9,7	6,8
1,1	3000	IES5	19,0	14,6	12,7	12,7	9,2	7,8	6,5	5,3
1,5	3000	IES5	14,7	11,6	10,2	9,7	7,3	6,1	5,5	4,2
2,2	3000	IES5	12,0	9,6	8,0	7,6	5,7	4,6	3,6	3,1
3	3000	IES5	10,9	8,7	7,7	7,0	4,9	8,2	3,2	2,7
4	3000	IES5	12,3	10,1	6,9	9,1	6,2	4,0	4,9	2,9
5,5	3000	IES5	10,6	7,4	6,3	6,2	4,1	3,6	2,8	2,3
7,5	3000	IES5	11,2	7,4	6,3	6,7	4,1	3,3	2,8	1,8

P_N	n	IES class	PL compared to the rated power [%]							
			Reference points							
[kW]	[rpm]		1	2	3	4	5	6	7	8
11	3000	IES5	9,3	5,5	4,7	4,7	3,4	2,6	1,4	1,6
15	3000	IES5	10,6	6,3	3,8	6,2	4,0	2,2	2,2	1,4
18,5	3000	IES5	8,0	5,0	3,4	5,2	3,0	2,0	2,1	1,4
22	3000	IES5	8,0	5,5	3,9	5,1	3,5	2,5	2,5	1,3
30	3000	IES5	8,1	5,7	3,9	4,9	3,3	2,3	2,5	1,3
37	3000	IES5	7,1	5,0	3,3	4,7	3,0	2,0	2,1	1,3
45	3000	IES5	7,6	5,1	3,6	4,8	3,1	2,2	2,1	1,4

Efficiency classes to IEC TS 60034-30-2:2016 and loss values to IEC TS 60034-2-3:2016

The motor losses indicated apply to operation on a KSB PumpDrive 2 / PumpDrive 2 Eco frequency inverter with the same rated power.

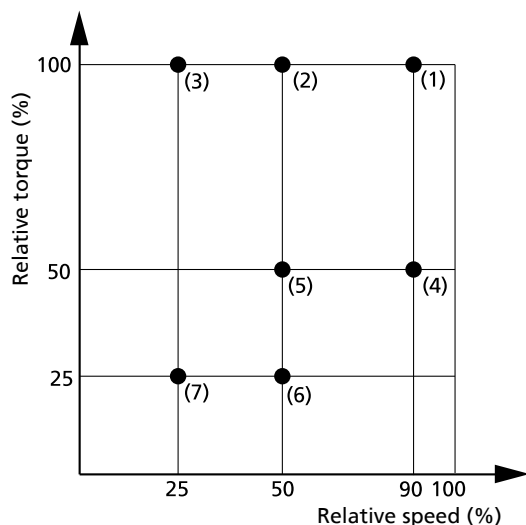


Fig. 2: Reference points to IEC TS 60034-2-3:2016

Table 12: Classification and loss values for reference points (1) to (7)

P_N	n	Efficiency class	Motor losses (W)						
			Reference points						
[kW]	[rpm]		1	2	3	4	5	6	7
0,55	1500	IE4	109	96	58	57	51	33	15
0,55	1500	IE5 ¹²⁾	87	66	60	28	20	11	7
0,75	1500	IE4	130	128	114	82	72	59	50
0,75	1500	IE5 ¹²⁾	104	67	61	32	23	12	8
1,1	1500	IE4	167	157	150	85	71	36	32
1,1	1500	IE5	134	126	120	68	57	29	26
1,5	1500	IE4	207	206	202	105	91	45	43
1,5	1500	IE5	165	164	161	84	73	36	34
2,2	1500	IE5	215	185	165	111	91	48	40
3	1500	IE5	267	263	243	141	117	65	56
4	1500	IE5	322	317	298	173	149	82	71
5,5	1500	IE4	502	538	437	245	245	104	94
5,5	1500	IE5	401	430	349	196	196	83	75
7,5	1500	IE5	492	464	417	226	197	83	80
11	1500	IE5	654	565	467	270	173	92	47
15	1500	IE5	801	589	524	320	241	63	122
18,5	1500	IE4	1177	834	712	513	374	225	187
22	1500	IE4	1331	899	735	599	453	265	178
30	1500	IE4	1662	1296	1086	837	642	364	268
37	1500	IE4	1938	1583	1198	928	690	382	294
45	1500	IE4	2267	1602	1326	1008	745	402	323
0,55	3000	IE5	104	66	54	52	37	24	16
0,75	3000	IE5	123	81	72	78	54	45	32
1,1	3000	IE5	158	120	100	96	69	45	32
1,5	3000	IE5	195	124	107	96	70	48	29
2,2	3000	IE5	247	142	119	101	72	40	30
3	3000	IE5	303	149	130	110	73	37	29
4	3000	IE5	369	280	205	257	166	137	90
5,5	3000	IE5	454	261	196	227	136	82	14
7,5	3000	IE5	555	346	270	305	179	108	64
11	3000	IE5	722	284	285	396	153	56	58

12 IE5 motors of sizes 0.55 kW / 0.75 kW with 1500 rpm are designed with permanent magnets; therefore a cogging torque is present when the motor is de-energised.

P_N	n	Efficiency class	Motor losses (W)						
			Reference points						
[kW]	[rpm]		1	2	3	4	5	6	7
15	3000	IE5	908	477	302	534	273	213	51
18,5	3000	IE5	1025	456	312	543	276	186	99
22	3000	IE4	1445	771	598	735	480	330	191
30	3000	IE4	1815	1036	803	946	577	425	230
37	3000	IE4	2088	1144	820	1130	654	448	299
45	3000	IE4	2448	1216	880	1289	698	449	255

Dimensions

Motor with adapter and PumpDrive 2

i Motor size 112 fitted with an additional adapter between motor housing and adapter plate for PumpDrive 2.

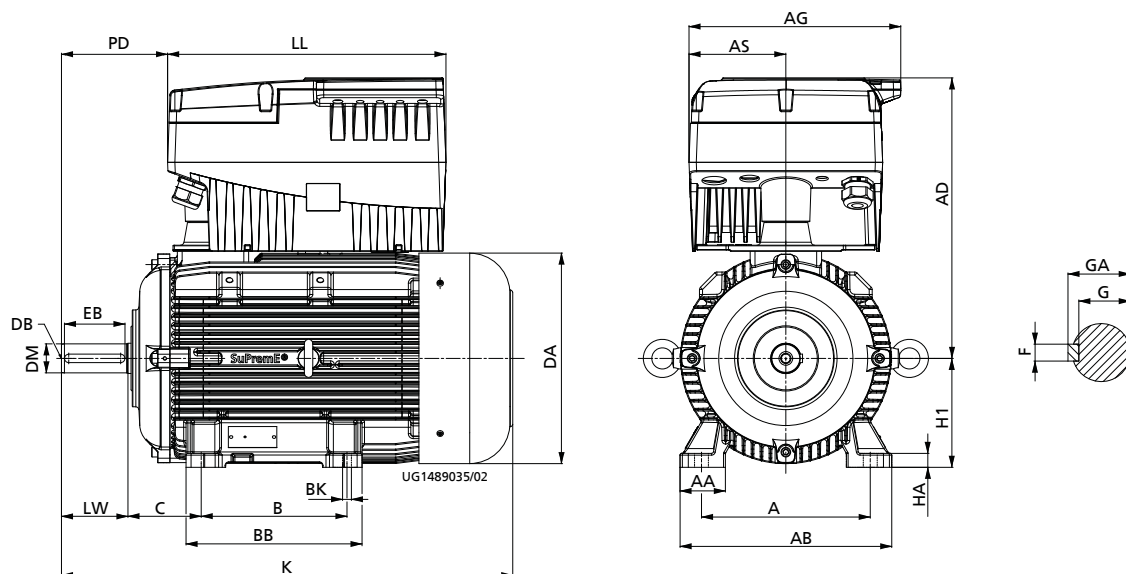


Fig. 3: Version with standard bearing cover (B3, etc.)

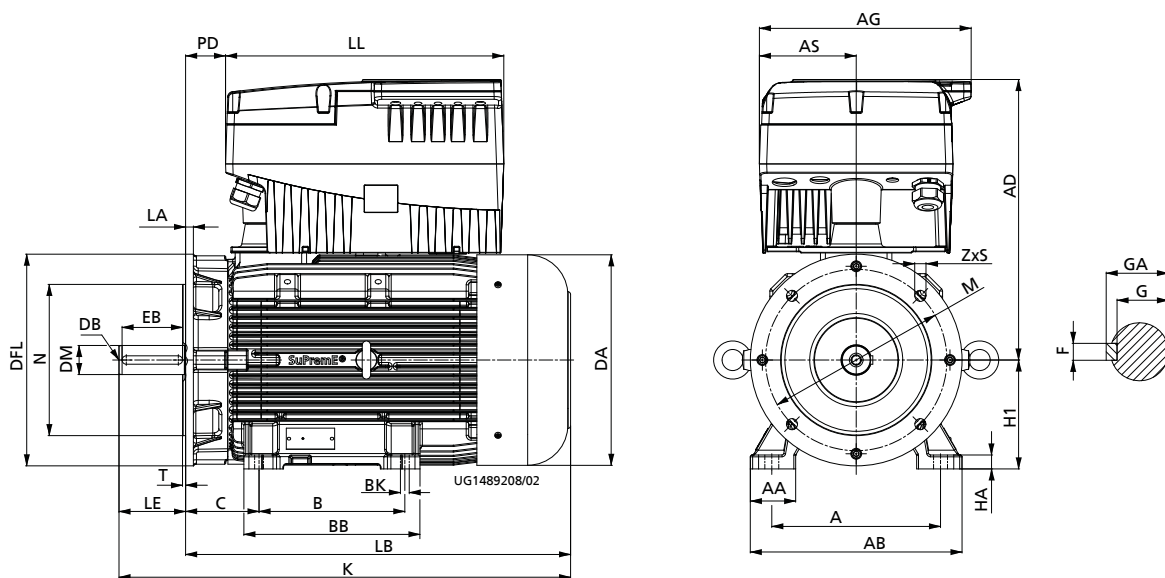


Fig. 4: Version with FF flange with clearance holes (V15, etc.)

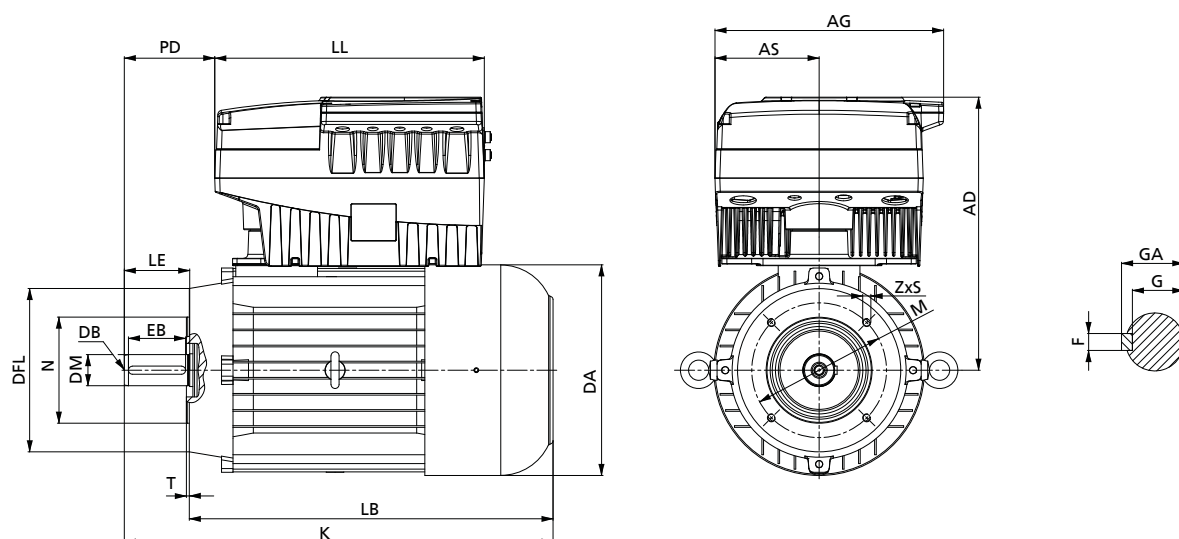


Fig. 5: Version with FT flange with tapped holes (V18, etc.)

Motor with terminal box

Table 13: Position of terminal box

Installation type	Shaft centreline height 71 - 160 mm	Shaft centreline height 180 - 225 mm
B3	Not defined	360°
V15	Not defined	360°
All installation types, with motor-mounted PumpDrive 2	360°	360°

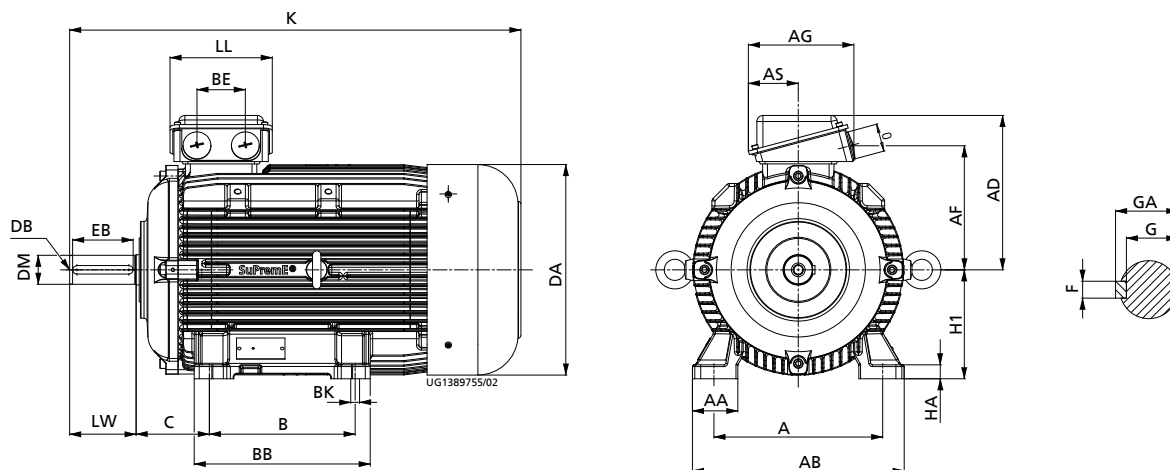


Fig. 6: Version with standard bearing cover (B3,...)

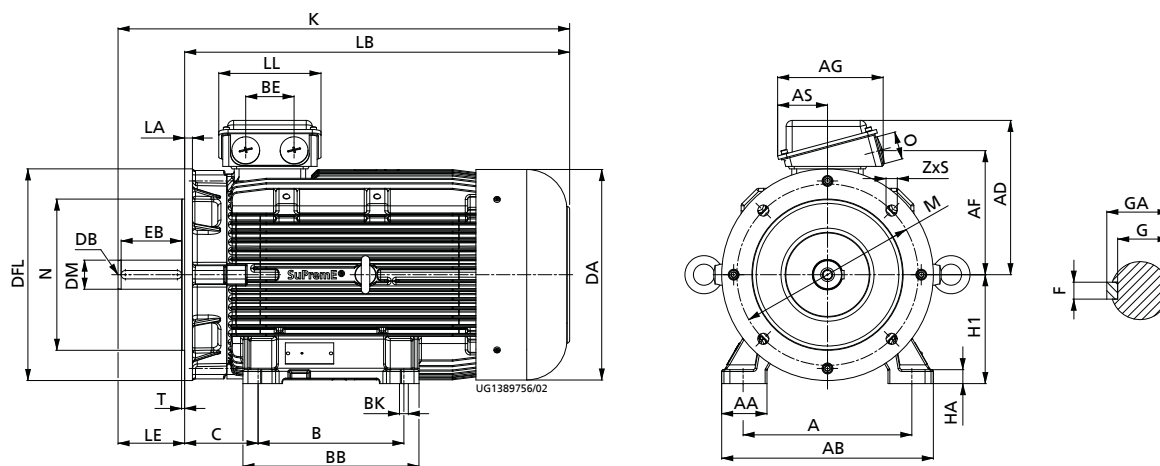


Fig. 7: Version with FF flange with clearance holes (V15,...)

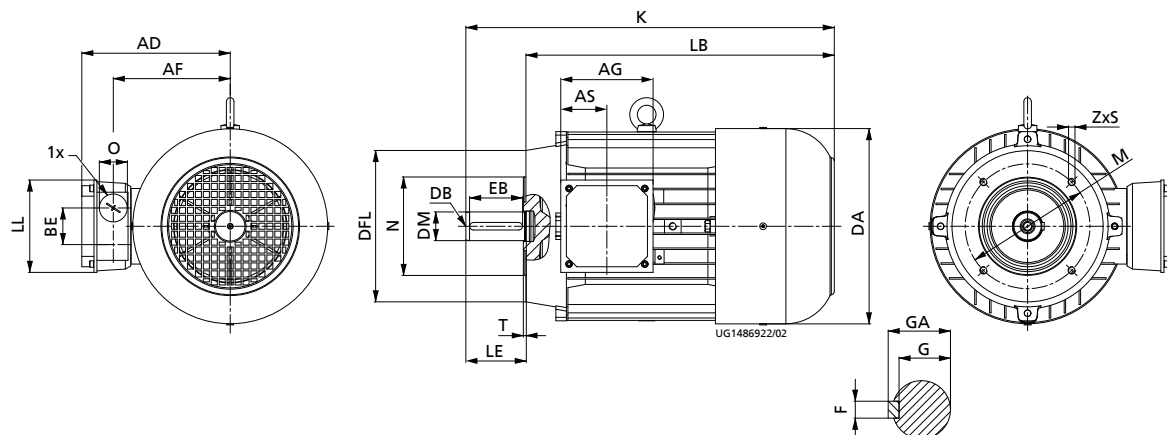


Fig. 8: Version with FT flange with tapped holes (V18,...)

Key

Table 14: Key: Code from dimension drawing = description

Code	Description	See drawing
A	Distance between centrelines of mounting holes	Front face
AA	Width of motor foot	Front face
AB	Distance from one outside edge to other outside edge of motor feet	Front face
AD	Distance between motor centreline and top edge of terminal box or PumpDrive	Front face
AF	Distance between motor centreline and cable gland opening	Front face
AG	Width of terminal box or PumpDrive	-
AS	Distance between motor centreline and outside edge on the side of terminal box or PumpDrive.	-
B	Distance between centrelines of mounting holes	Side view
BB	Distance from one outside edge to other outside edge of motor feet	Side view
BE	Distance between centrelines of the two cable gland openings	Side view
BK	Diameter of mounting holes in motor foot	Side view
C	Distance between shaft shoulder and centreline of mounting holes of adjacent motor feet	Side view
DA	Motor diameter	Front face
DB	Thread size of centre hole	Side view
DFL	Flange diameter	Side view
DM	Shaft diameter	Side view
EB	Length of drive-end key	Side view
F	Width of drive-end key	Detail
G	Distance between keyway base and opposite surface of the drive-end shaft end	Detail
GA	Distance between top of the key and opposite surface of the drive-end shaft end	Detail
H1	Shaft centreline height	Front face
HA	Motor foot thickness	Front face
K	Overall length of motor (incl. shaft end)	Side view
LA	Flange width	Side view
LB	Distance from flange mounting face to motor end (without shaft stub)	Side view
LE/LW	Length of shaft stub ¹³⁾	Side view
LL	Width of terminal box or PumpDrive	Side view
M	Diameter of mounting bolt circle at flange	Front face
N	Flange centring diameter	Side view
O	Diameter of cable gland opening	Side view or front face
PD	Distance between shaft end or flange mounting face to outer edge of PumpDrive	-
S	Flange bolt hole diameter or nominal thread size	Front face
T	Flange centring depth	Side view
Z	Number of mounting holes	-

13 LE = for installation types V1, V15 and V18, LW = installation type B3

Motor

Table 15: Motor dimensions

Motor	DA	DB	DM	H1	K	LE/LW ¹⁴⁾
	[mm]					
71M	138	M5	14	71	265	30
80M	158	M6	19	80	292	40
90S	176	M8	24	90	337	50
90L	176	M8	24	90	361	50
100L	194	M10	28	100	400	60
112M	219	M10	28	112	425	60
132S	259	M12	38	132	481	80
132M	259	M12	38	132	519	80
160M	308	M16	42	160	613	110
160L	308	M16	42	160	658	110
180M	356	M16	48	180	745	110
180L	356	M16	48	180	773	110
200L	396	M20	55	200	846	110
225S	449	M20	60	225	905	140
225M (2 poles)	449	M20	55	225	882	110
225M (4 poles)	449	M20	60	225	928	140

Motor feet

Table 16: Motor feet dimensions

Motor	A	AA _{max}	AB _{max}	B	BB _{max}	BK	C	HA _{max}
	[mm]							
71M	112	30	140	90	108	7	45	11
80M	125	36	160	100	125	10	50	13
90S	140	41	180	100	132	10	56	13
90L	140	41	180	125	155	10	56	12
100L	160	47	200	140	173	12	63	13
112M	190	51	228	140	177	12	70	14
132S	216	60	260	140	180	12	89	18
132M	216	60	260	178	218	12	89	18
160M	254	72	318	210	264	14,5	108	22
160L	254	72	318	254	308	14,5	108	22
180M	279	75	350	241	328	15	121	23
180L	279	75	350	279	328	15	121	23
200L	318	71	380	305	365	19	133	25
225S	356	92	436	286	361	19	149	34
225M	356	92	436	311	406	19	149	34

14 LW = installation type B3

Terminal box

Table 17: Terminal box dimensions

Motor	AD	AF	AG	AS	BE	LL	O
	[mm]						
71M	127	90	114	57	57	153	M25
80M	131	99	114	57	47	114	M25
90S	137	106	114	57	47	114	M25
90L	137	106	114	57	47	114	M25
100L	149	114	114	57	47	114	M25
112M	169	133	120	60	50	142	M32
132S	191	154	122	61	50	142	M32
132M	191	154	122	61	50	142	M32
160M	239	189	186	93	83	186	M40
160L	239	189	186	93	83	186	M40
180M	296	239	265	112	85	197	M40
180L	296	239	265	112	85	197	M40
200L [30 kW]	314	257	265	112	90	224	M50
200L [37 kW]	314	257	265	112	90	224	M50
225S	339	282	266	124	90	224	M50
225M	339	282	266	124	90	224	M50

Motor-mounted PumpDrive 2

Table 18: Dimensions of motor-mounted PumpDrive 2

Motor	Size PumpDrive 2	AD	AG	AS	LL	PD	
						B3	V15
		[mm]					
71M	A	237	190	85	260	35	4,5
80M	A	258	190	85	260	45	4,5
90S	A	265	190	85	260	58	8
90L [1,5 kW]	A	265	190	85	260	58	8
90L [2,2 kW]	B	255	211	93	290	63	11
100L	B	267	211	93	290	77	17
112M	B	315	211	93	290	89	29
132S	C	340	280	128	330	110	33
132M	C	340	280	128	330	110	33
160M [11 kW]	C	369	280	128	330	156	49
160M [15 kW]	D	458	350	160	460	169	62
160L	D	458	350	160	460	169	62
180M	D	463	350	160	460	175	80
180L	D	463	350	160	460	175	80
200L [30 kW]	D	480	350	160	460	205	89
200L [37 kW]	E	535	455	213	700	187	88
225S	E	556	455	213	700	233	88
225M	E	556	455	213	700	203	88

Key

Table 19: Key dimensions

Motor	EB	F	G	GA
	[mm]			
71M	22	5	11	16
80M	32	6	15,5	21,5
90S	40	8	20	27
90L	40	8	20	27
100L	50	8	24	31
112M	50	8	24	31
132S	70	10	33	41
132M	70	10	33	41
160M	90	12	37	45
160L	90	12	37	45
180M	100	14	42,5	51,5
180L	100	14	72,5	51,5
200L	100	16	49	59
225S (2 poles)	125	18	53	64
225M (2 poles)	100	16	49	59
225M (4 poles)	125	18	53	64

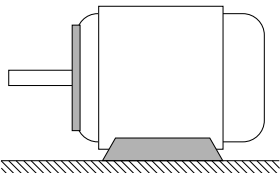
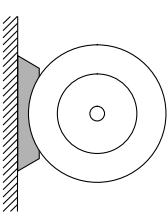
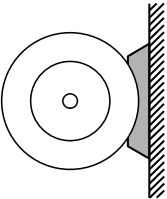
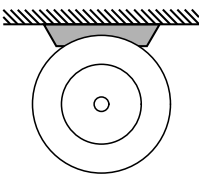
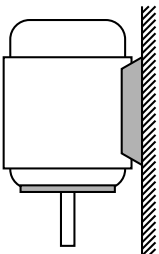
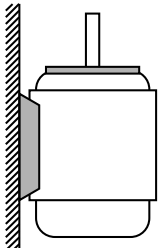
Motor flange

Table 20: Motor flange dimensions

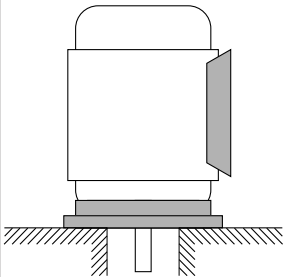
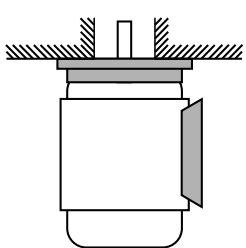
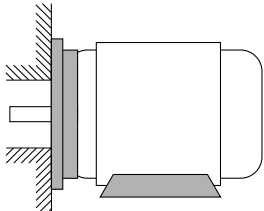
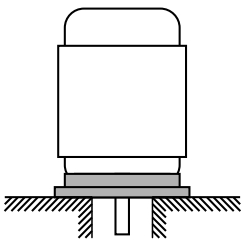
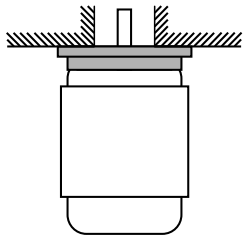
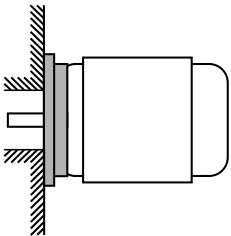
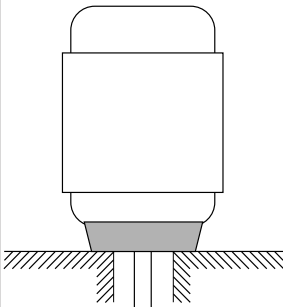
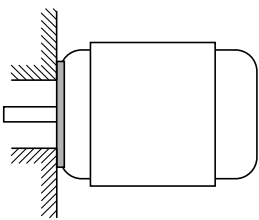
Motor	Flange type														
	FT	FF	FF	FT	FF	FT	FF	FT	FF	FT	FF	FT	FF	FT	FF
	DFL		LA _{max}	LB _{max}		M		N		S		T		Z	
	[mm]														
71M	105	160	10	214	214	85	130	70	110	M6	10	2,5	3,5	4	4
80M	120	200	10	252	252	100	165	80	130	M6	12	3	3,5	4	4
90S	140	200	10	287	287	115	165	95	130	M8	12	3	3,5	4	4
90L	140	200	10	311	311	115	165	95	130	M8	12	3	3,5	4	4
100L	160	250	15	340	340	130	215	110	180	M8	14,5	3,5	4	4	4
112M	160	250	15	365	365	130	215	110	180	M8	14,5	3,5	4	4	4
132S	200	300	16	401	401	165	265	130	230	M10	14,5	3,5	4	4	4
132M	200	300	16	439	439	165	265	130	230	M10	14,5	3,5	4	4	4
160M	250	350	24	503	503	215	300	180	250	M12	18,5	4	5	4	4
160L	250	350	24	548	548	215	300	180	250	M12	18,5	4	5	4	4
180M	-	350	13	635	635	-	300	-	250	-	18,5	-	5	4	4
180L	-	350	13	663	663	-	300	-	250	-	18,5	-	5	4	4
200L	-	400	15	736	736	-	350	-	300	-	18,5	-	5	4	4
225S	-	450	16	747	747	-	400	-	350	-	18,5	-	5	8	8
225M	-	450	16	788	788	-	400	-	350	-	18,5	-	5	8	8

Mounting arrangements

Table 21: Mounting arrangements

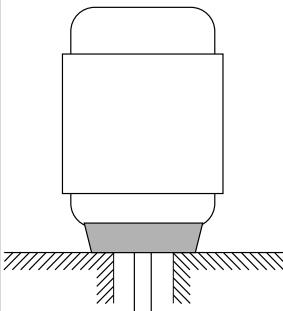
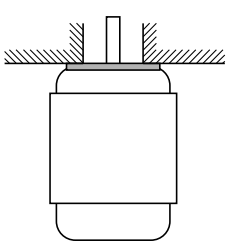
Standard mounting arrangement	Further mounting arrangements	Shaft centreline height		
		H ≤ 132	H ≤ 160	H ≥ 180
		[mm]		
 IM B3	 IM B6	-	✓	✓
	 IM B7	-	✓	✓
	 IM B8	-	✓	✓
	 IM V5	-	✓	✓
	 IM V6	-	✓	✓

4075.53/16-EN

Standard mounting arrangement	Further mounting arrangements	Shaft centreline height		
		H ≤ 132	H ≤ 160	H ≥ 180
		[mm]		
IM V15 	IM V35 	-	✓	✓
	IM B35 	-	✓	✓
IM V1 	IM V3 	-	✓ ¹⁵⁾	-
	IM B5 	-	✓ ¹⁵⁾	-
IM V18 	IM B14 	✓	-	-

4075.53/16-EN

15 Feet can be removed from IM V15 basic type of construction

Standard mounting arrangement	Further mounting arrangements	Shaft centreline height		
		H ≤ 132	H ≤ 160	H ≥ 180
		[mm]		
IM V18 	IM V19 	✓	-	-

Available torque

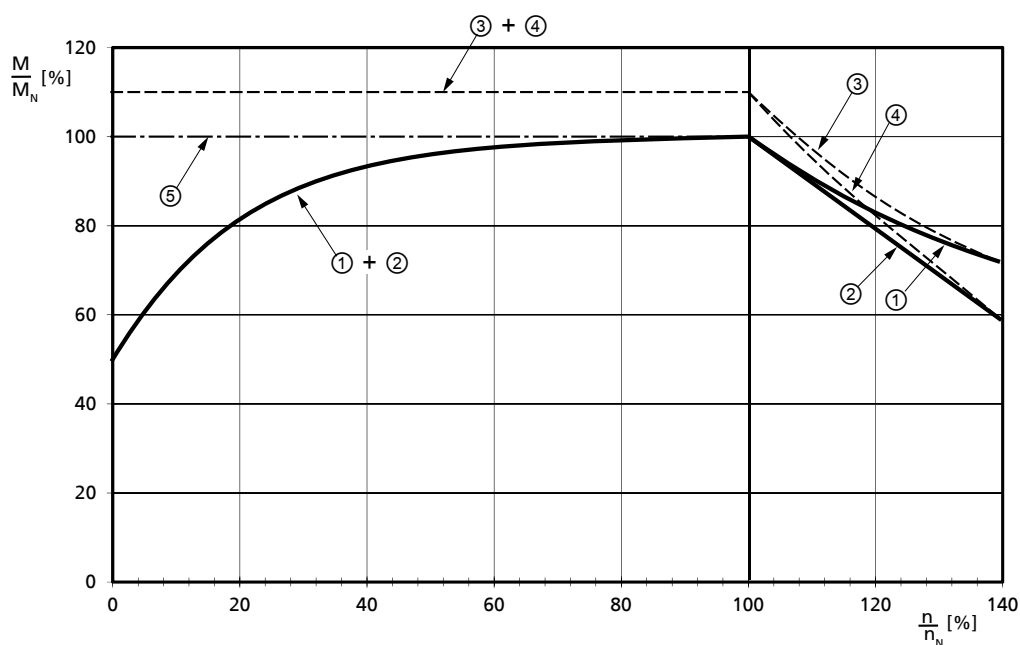


Fig. 9: Available torque

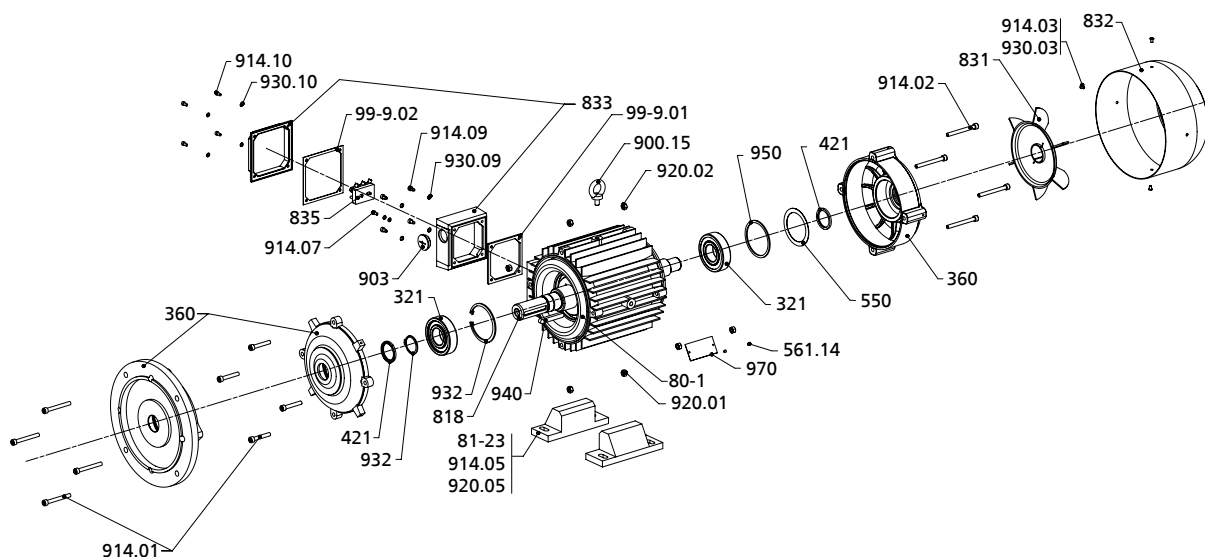
①	IC 411	S1 continuous duty
②	IC 411, from size 180	S1 continuous duty
③	IC 411 / IC 416	S2 duty (max. 10 seconds)
④	IC 411 / IC 416, from size 180	S2 duty (max. 10 seconds)
⑤	IC 416	S1 continuous duty

IC411 For surface cooling with fan mounted on drive shaft/TEFC (DIN EN 60034-6)

IC416 For external cooling by separate fan/TEFV (DIN EN 60034-6)

Cross sectional drawings with list of components

Variant made of aluminium



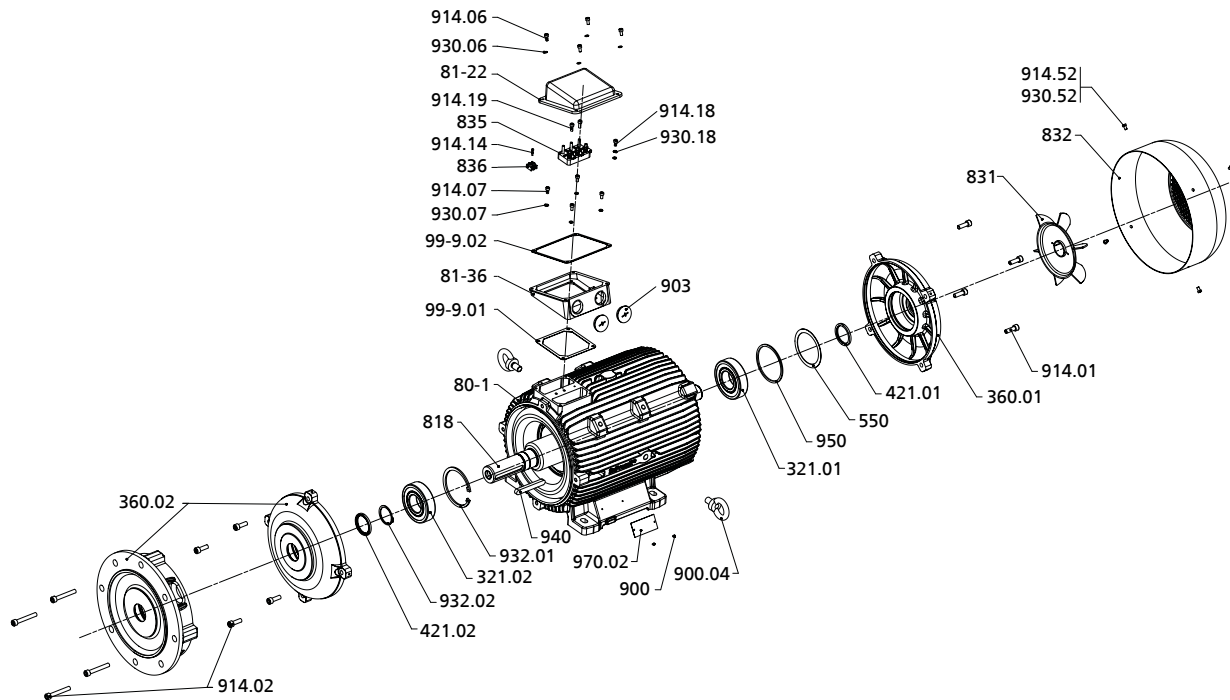
UG1492721

Fig. 10: Variant made of aluminium

Table 22: List of components

Part No.	Description	Part No.	Description
80-1	Motor unit (motor housing with stator)	833	Terminal box
81-23	Motor foot	835	Terminal board
99-9.01/02	Set of sealing elements	900.15	Bolt/screw
321	Radial ball bearing	903	Screw plug
360	Bearing cover	914.01/02/03/05/07/09/10	Hexagon socket head cap screw
421	Lip seal	920.01/02/05	Nut
550	Disc	930.03/09/10	Safety device
561.14	Grooved pin	932	Circlip
818	Rotor	940	Key
831	Fan impeller	950	Spring
832	Fan hood	970	Name plate

Variant made of grey cast iron



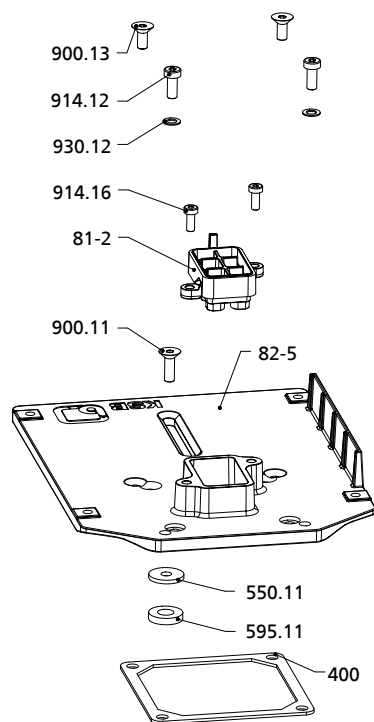
UG1492278

Fig. 11: Variant made of grey cast iron

Table 23: List of components

Part No.	Description	Part No.	Description
80-1	Motor unit (motor housing with stator)	835	Terminal board
81-22	Terminal box cover	836	Terminal strip
81-36	Terminal box base	900.04	Bolt/screw
99-9.01/.02	Set of sealing elements	903	Screw plug
321.01/.02	Radial ball bearing	914.01/.02/.06/.07/.14/.18/.19/.52	Hexagon socket head cap screw
360.01/.02	Bearing cover	930.06/.07/.18/.52	Safety device
421.01/.02	Lip seal	932.01/.02	Circlip
550	Disc	940	Key
818	Rotor	950	Spring
831	Fan impeller	970.02	Name plate
832	Fan hood		

Adapter, sizes A and B



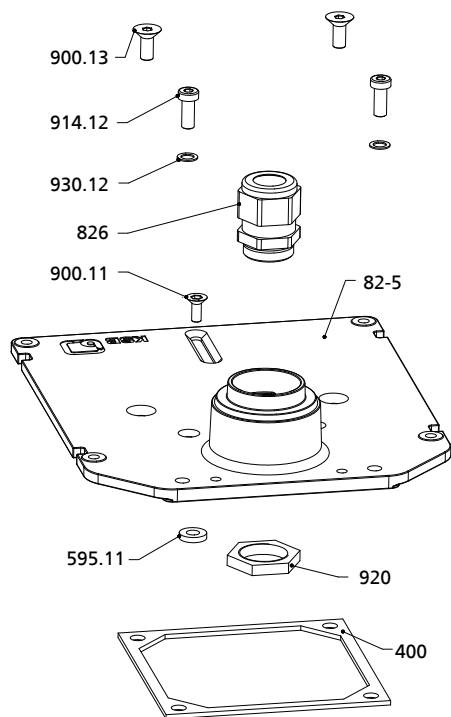
UG1492488

Fig. 12: Adapter, sizes A and B

Table 24: List of components

Part No.	Description	Part No.	Description
81-2	Plug	595.11	Anti-vibration pad
82-5	Adapter	900.11/.13	Bolt/screw
400	Gasket	914.12/.16	Hexagon socket head cap screw
550.11	Disc	930.12	Safety device

Adapter, size C



UG1492488

Fig. 13: Adapter, size C

Table 25: List of components

Part No.	Description	Part No.	Description
82-5	Adapter	900.11/.13	Bolt/screw
400	Gasket	914.12	Hexagon socket head cap screw
595.11	Anti-vibration pad	920	Nut
826	Cable gland	930.12	Safety device

Adapter, sizes D and E

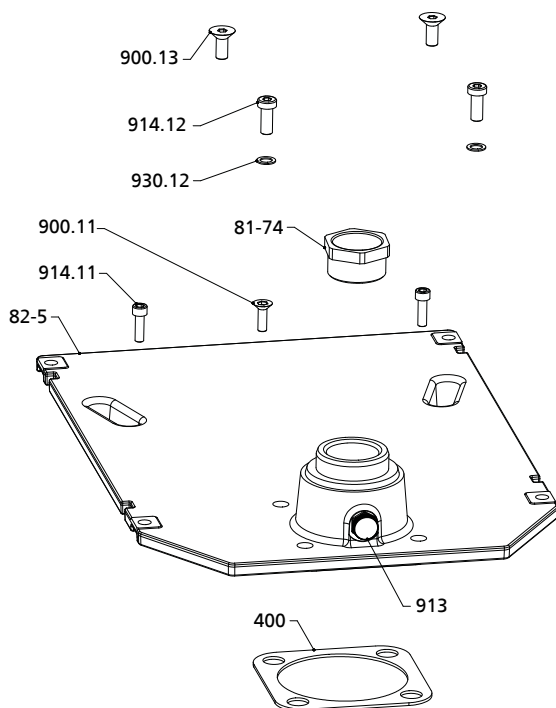


Fig. 14: Adapter, sizes D and E

Table 26: List of components

Part No.	Description	Part No.	Description
81-74 ¹⁶⁾	Pressure screw	913	Pressure balancing element
82-5	Adapter	914.11/.12	Hexagon socket head cap screw
400 ¹⁷⁾	Gasket	930.12	Safety device
900.11/.13 ¹⁶⁾	Bolt/screw		

16 Not when using an adapter with pressure balancing element
17 Only when using an adapter with pressure balancing element

Glossary

IE4

Efficiency class to IEC TS 60034-30-2:2016 = Super Premium Efficiency (IE = International Efficiency)

IE5

Efficiency class for rotating electrical machinery to IEC TS 60034-30-2:2021 = Ultra Premium Efficiency (IE = International Efficiency)

Mat. No.

This identification number is composed of an 8-digit numerical code that uniquely identifies a product entered in SAP.

